

CSCN8020 Assignment 3 Submission

Deep Q-Network Control of the Unitree G1 Left Elbow

Student full name	Haibo Yuan
Student ID	9010929
Assignment title	CSCN8020 Assignment 3 - Unitree G1 DQN Left Elbow Control

Project Summary

This project implements a student-written PyTorch Deep Q-Network for the approved Unitree G1 left elbow Gymnasium environment. The DQN maps the elbow angle, velocity, target angle, and angle error to three discrete action values: decrease, hold, or increase the controller target. Two epsilon-decay settings were trained headlessly on CPU and evaluated greedily across four benchmark goals. The selected checkpoint reached 20/20 successful final evaluation episodes and was compared with the provided rule-based policy using saved metrics, plots, a technical report, and a rendered MuJoCo demonstration video.

GitHub Repository URL

https://github.com/hyuan0929/CSCN8020_Assignment3

Cloneable .git URL

https://github.com/hyuan0929/CSCN8020_Assignment3.git

Submission Notes

The repository contains the completed notebook, student-written DQN source files, saved checkpoint, metrics, plots, README commands, and the full technical report. The rendered video is submitted separately or by accessible link according to Brightspace instructions.